



ENVT. MGT. Department
SIMHADRI

Note No :1



Ref:SMTPP/ENVT. MGT./2024-25/EMG-AU/392728

Date:09/08/2024(15:49:53)

Subject:LMI on Monitoring of Ambient Air Quality

Competent authority had accorded approval vide notesheet :SMTPP/ENVT. MGT./2022-23/EMG-AU/304201 dated 25.07.2022(Copy enclosed) for the approval of LMI - Monitoring of ambient Air quality Rev07.

However a revised OGN is issued with following revisions on 5th August 2024.

Records of revision in OGN

- LMI reviewing Authority has been updated to HOP.
- Addition of Annexure I which is a Checklist for Ambient Air Monitoring and Annexure V for Meteorological data format Weather Phenomenon.
- Record of Deviation % to be recorded in the last sampling for validation/ verification.
- Mentioning of reasons for less availability (<85%) of AAQMS data in Annexure IV.

In addition to that there has been an increase in the number of CAAQMS stations with addition of 3 more CCAQMS stations in our project.

Accordingly, the LMI has been revised and put up for review and approval please.

Submitted for approval please.

Submitted by:

सिबिन अब्राहम - वरिष्ठ प्रबंधक

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On:09/08/2024(15:49:53) **Ref:**SMTTPP/ENVT. MGT./2024-25/EMG-AU/392728

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On:10/08/2024(15:38:06) **Ref:**SMTTPP/ENVT. MGT./2024-25/EMG-AU/392728

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On:10/08/2024(15:49:30) **Ref:**SMTTPP/ENVT. MGT./2024-25/EMG-AU/392728

Note No:4

LMI document is submitted for review and approval.

Submitted by:

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On:12/08/2024(16:34:06) **Ref:**SMTTPP/ENVT. MGT./2024-25/EMG-AU/392728

Note No:7

Please review.

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Note No:8

Document is reviewed and found ok.

Submitted by:

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Note No:9

May pl see the revised LMI

Submitted by:

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On:20/08/2024(10:54:34) **Ref:**SMTTP/ENV.T. MGT./2024-25/EMG-AU/392728

Note No:10

Document is reviewed and found ok

Submitted by:

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On:28/08/2024(11:48:20) **Ref:**SMTTP/ENV.T. MGT./2024-25/EMG-AU/392728

Note No:11

LMI is forwarded for approval

Submitted by:

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On:28/08/2024(12:18:13) **Ref:**SMTTP/ENV.T. MGT./2024-25/EMG-AU/392728

Note No:12

Revisions proposed in the LMI may be recorded in the proposal

Submitted by:

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On:30/08/2024(15:18:21) **Ref:**SMTTP/ENV.T. MGT./2024-25/EMG-AU/392728

Note No:13

Submitted by:

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On:30/08/2024(17:58:54) **Ref:**SMTTP/ENV.T. MGT./2024-25/EMG-AU/392728

Note No:14

Following are the revisions proposed in the LMI through a revised OGN

- LMI reviewing authority has been updated to HOP from HOD (EMG-AU).
- Addition of Annexure I which is a checklist for Ambient Air Monitoring.
- Addition of Annexure V for Meteorological data format Weather Phenomenon.
- Record of Deviation % to be recorded in the last sampling for validation/verification in Annexure IV.
- Mentioning of reasons for less availability (<85%) of AAQMS data in Annexure IV.

In addition to that there has been an increase in the number CAAQMS stations with an addition of 3 more CAAQMS stations for PM 10 monitoring in our project.

- Dasaripetta.
- Devada.
- Pittavanipalem.

Accordingly, the LMI has been revised and put up for review and approval please.

Submitted for approval please.

Submitted by:

सिबिन अब्राहम - वरिष्ठ प्रबंधक

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On:02/09/2024(11:19:56) **Ref:**SMTTPP/ENVT. MGT./2024-25/EMG-AU/392728

Note No:15

Submitted for review and further approval.

Submitted by:

डॉ वी जयन - अपर महाप्रबंधक

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On:03/09/2024(18:01:02) **Ref:**SMTTP/ENVT. MGT./2024-25/EMG-AU/392728

Note No:16

May be approved

Submitted by:

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On:07/09/2024(11:37:20) **Ref:**SMTTP/ENVT. MGT./2024-25/EMG-AU/392728

Note No:17

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On:09/09/2024(18:42:53) **Ref:**SMTTP/ENVT. MGT./2024-25/EMG-AU/392728

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

Monitoring of Ambient Air Quality

LMI/OGN/ENV/010 Rev: 08

AUGUST -2024

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	Monitoring of Ambient Air Quality
2	LMI No.	LMI/OGN/ENV/010 Rev:08
3	Reference Document	COS-ISO-00-OGN/ENV/010/Rev 03 August 2024
4	Superseded Document	LMI/OGN/ENV/010 Rev:07 JUNE 2022
5	Prepared By	Sibin Abraham, Sr. Manager (EMG-AU)
6	Checked By	Dr V. Jayan, AGM (EMG& AU)
		HOD (Chem)
		HOD (C&I)
		GM(O&M)
7	Reviewed & Approved By	S. K. Sinha HOP (Simhadri)

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TITLE	Monitoring of Ambient Air quality
LMI No.	LMI/OGN/ENV/010 Rev:08

The purpose of the LMI on Monitoring Ambient Air Quality are as follows:

- Monitoring of ambient air quality at locations inside or outside plant area and reporting as per the NAAQS -2019 monitoring criteria.
- Air Quality Management to assess the present status of air quality and effectiveness of air pollution control strategies and long-term management of air quality in the plant area.
- To determine the requirement of additional air pollution control measures, if any.
- To identify specific industrial and other sources of ambient air pollution.
- To collect data for formulating and testing air pollution models.

2.0 Superseded Documents

LMI/OGN/ENV/010 REV. 7

3.0 Details of Revision

Following are the proposed revisions

- Addition of Annexure I as per the latest approved OGN.
- Increase in the number of CAAQMS stations.
- LMI approval authority of HOP addition.
- Record of Deviation % to be recorded in the last sampling for validation/ verification.
- Mentioning of reasons for less availability (<85%) of AAQMS data in Annexure III.

4.0 Identification of LMI

LMI/OGN/ENV/010 Rev 08

5.0 Reference Documents

TITLE	Monitoring of Ambient Air quality
LMI No.	LMI/OGN/ENV/010 Rev:08

- Environmental clearance from MoEF & CC.
- Consent orders of APPCB.
- Guidelines for the Measurement of Ambient Air Pollutants:
 - Volume -I: Guidelines for manual sampling and analyses (along with sample flow chart and data sheets).
 - Volume-II: Guidelines for continuous sampling and real time analyses.
- The Environmental (Protection) Act 1986.
- The Air (Prevention and control of Pollution) Act 1981
- Office Memorandum dated 19.11.2018, MoEF & CC.

6.0 The Framework of Ambient Air Quality Monitoring

- **Environment Management**

Effective environmental management depends on the availability of organized information on the state of environment and its relationship to social and economic factors. It is therefore, essential that reliable environmental information is generated and the same maintained in a usable form.

The process of generation of information may be termed as environment monitoring.

An environmental monitoring program comprises of distinct elements

- a) Installation of continuous ambient air quality monitoring station (CAAQMS).
- b) Monitoring Network.
- c) Laboratory Support (Manual sampling & analysis) when required.

- **Monitoring Network**

Design of Monitoring Network for Ambient Air Quality Monitoring, is basically governed by the factors outlined below: -

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▪ **Standards for Ambient Air Quality**

Central Pollution Control Board (CPCB) under Environmental Protection Rules has specified National Ambient Air Quality standards as per notification issued on 18th November 2009. The National Ambient Air Quality Standards as per the CPCB Notification is enclosed for reference – as Annexure - II.

Standards for other parameters as mentioned in the National Ambient Air Quality Standards CPCB Notification No.B-29016/20/90/PCI-I, dated 18.11.2009.

The consent order issued by APPCB specified the standards for ambient air quality as follows.

Sl. No	Parameter	Maximum Limit (Daily average)
01	Respirable Particulate matter (PM10)	100 µgm/M ³
02	Sulphur dioxide	80 µgm/M ³
03	Oxides of Nitrogen	80 µgm/ M ³
04	Particulate Matter (PM2.5)	60 µgm/M ³

▪ **Offline and online monitoring:**

For monitoring of ambient air quality in plant and township area, NTPC Simhadri is equipped with state of art online ambient air quality monitoring stations in adequate numbers as stipulated by MoEF&CC/APCB and data is available on continuous basis for APCB/CPCB. Offline monitoring is also being carried out as per conditions stipulated by MoEF&CC/CPCB/APCB's. As offline monitoring stations are not fixed it provides flexibility of reach to specific locations for stipulated conditions or as required, validation/verification of online data as well as system checks of AAQ monitoring and analysis as per standards.

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▪ **Selection of Sampling Procedure :-**

For determination of concentration of a gaseous pollutant in air, it is necessary to obtain representative samples of air at a number of strategic points. The concentration is likely to vary with time and distance from the source of pollution depending upon the nature of release and local conditions such as topography and presence of buildings, other tall structures or vegetation. These aspects are required to be taken into account for sampling and interpretation of the measured values.

The samples thus obtained may be analyzed by established physical, chemical methods to determine their composition.

The selection of sampling procedure depends upon the nature of pollutants, the expected range of their concentrations, and the relevant time average needed.

▪ **Location of Air Quality Monitoring Stations:-**

Location of Air quality monitoring stations is based on computer modeling done by Corp. Env. Eng. Group. The details are as follows.

Location	Direction	Distance (in kms)	Basis of selection
AQ1	North East	4-5 Km	Maximum impact
AQ2	East/West	3-4 Km	2 nd Maximum impact
AQ3	South East	2-3 Km	3 rd Maximum impact
AQ4	South	5 Km	No impact (pollution Free zone)
AQ5	South West	<3km	4 th Maximum Impact

Based on above and aspects like housing the equipment, availability of power, security, accessibility to the location throughout the year etc.,

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three

no. online/continuous ambient air quality stations are set up at

1. BBPS, Deepanjalinagar
2. R&D Building
3. Kalapakka

In addition to that 3 more online continuous ambient air quality are set up for PM 10 monitoring at

1. Dasaripetta
2. Devada
3. Pittavanipalem

As per AP Pollution Control Board (APPCB) Consent, NTPC-Simhadri has to maintain 3 CAAQMS stations connected to APPCB website. In addition, to meet the other requirements the number of locations can be increased.

▪ **Number of Sampling Stations & Measurements**

Considering the above and requirements of PCB, locations are selected for monitoring the ambient air quality through manual measurements.

1. Kalapakka
2. Devada
3. Rawada village
4. NTPC township
5. R&D Building
6. Pittavanipalem

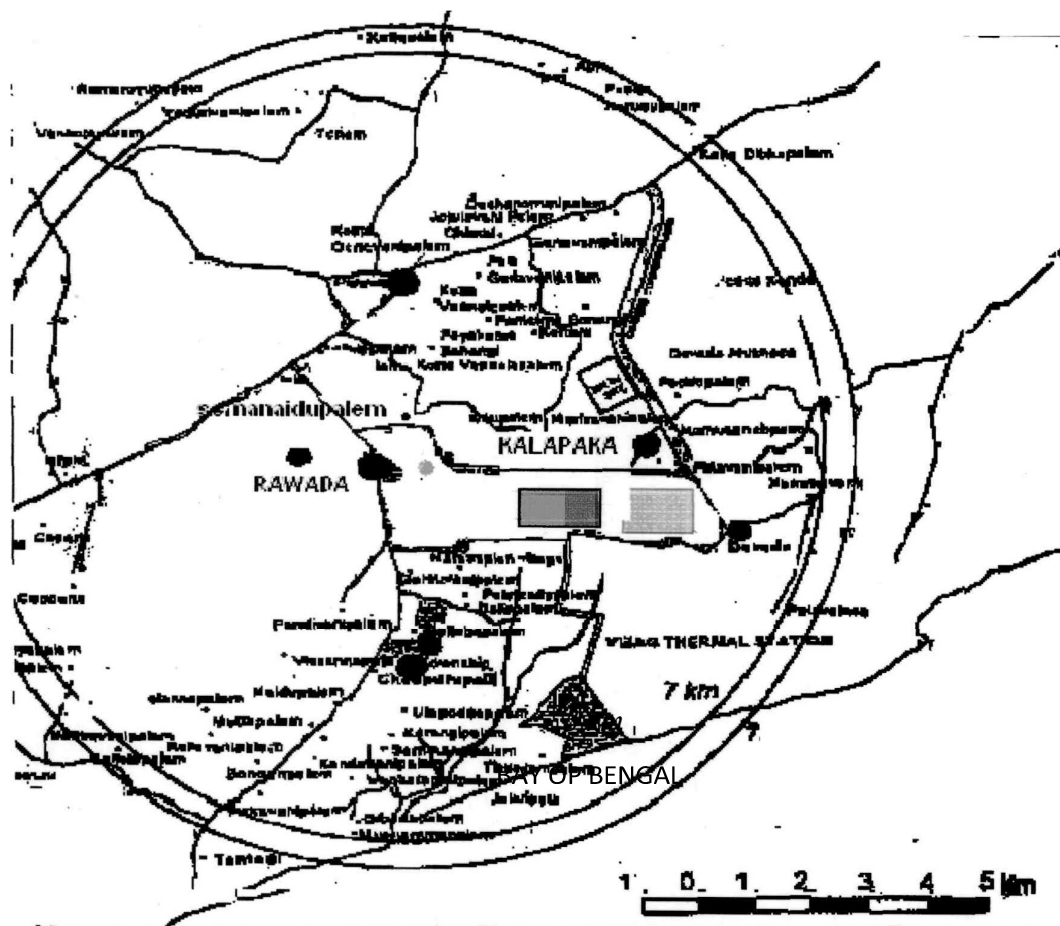
TITLE	Monitoring of Ambient Air quality
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7. Dasaripeta

The monitoring locations are as per the map shown in Fig. 1

As per AP Pollution Control Board Consent, Respirable Particulate Matter (PM₁₀), Particulate Matter (PM_{2.5}), Sulfur dioxide (SO₂), Nitrogen oxide (NO_x) are to be monitored in Ambient Air.

Fig .1 Ambient air quality monitoring stations locations



- Laboratory Support (Manual Sampling and Analysis)

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Laboratory support is essential for sample collection, sample preservation and analysis. The period, frequency and duration of sampling should be appropriate to the objective of the study. The meteorological and source emission factors, which influence the atmospheric concentration, should be taken into account. The consent order issued by APPCB and the guidelines issued by CPCB to be followed to specify the parameters, frequency and duration of testing etc.

The most important part of the monitoring program is compiling the monitored data in an easily retrievable and interpretable form. Annexure IV gives the format in which data for ambient air quality is to be recorded.

Specific instructions for sampling and analysis:

- Sample shall be collected for a duration of 24 hours and once in a month at locations BBPS,R&D,Kalpaka for 4 parameters (PM10,PM2.5,SO2, NOx) and other 8 parameters (Ozone O3, Lead Pb, Carbon monoxide (CO),Ammonia (NH3), Benzene(C6H6), Benzopyrene(BaP), Arsenic (As) and Nickel(Ni)) to be tested once in six months.
 - Total suspended matter shall be collected in high volume sampler with average gas flow rate not less than 1.1 cubic meter / minute. Respirable particulate matter shall be collected on respirable particulate matter sampler.
 - Sample for sulphur dioxide shall be collected on 24 hr basis and analysis method will be improved West & Geake method or ultra violet fluorescence .
 - Sample for oxides of nitrogen shall be collected on 24 hr basis and analysis method will be Jacob & Hochheiser Modified (Na-Arsenite) method or gas phase chemiluminescence.
-
- **Calibration of Continuous Ambient Air Quality Monitoring Station (CAAQMS) :**

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Calibration of the CAAQMS shall be done on monthly basis as per the procedure mentioned in O&M manual by OEM and record of the same shall be maintained in logbook and displayed as well at respective CAAQMS. Calibration gas validity certificate issued by the supplier or issuing authority shall be keep available at all the time for all type of calibration gases. Validation of online data to be done jointly with C&I, Chem and EMG with help of manual sampling for all monitored parameters and % deviation to be recorded. Accordingly, calibration correction of analyser and corrective action to be taken up by C&I. Calibration details and deviation to be recorded in format as per Annexure-3. The availability of CAAQMS shall be ensured 24×7 without failure except inevitable circumstances. Record of details on calibration, CAAQMS availability and exception if any shall be communicated to Head (EMG), RHQ and Head (EMG) CC on monthly basis.

7.0 Roles and Accountability

- Ensure Ambient Air Quality Monitoring as per National Ambient Air Quality Standards-2009 (Annexure-II) by offline/online monitoring system as per APCB/CPCB protocols (Annexure-III) and for data transmission to designated regulatory agencies from CAAQMS and as per applicable amendment of existing standards/new notifications by APCB and other Regulatory bodies.

Res. HOP(Simhadri)

- At R&D Building, NTPC Township & Kalapaka continuous Ambient air quality monitoring stations are installed along with PM 10 monitoring stations at Devada, Dasaripetta and Pittavanipalem. The monthly calibration of these equipment is to be carried out.

Res. HOD (C&I)

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- The data from the above stations is to be captured at Central Control Room and is to be preserved for future records for onward transmission to Statutory Authorities as per the requirement.

Res. HOD (EMG)

- At Pittavanipalem, Dasaripeta, Devada and Rawada, the off-line measurements are to be carried out through MOEF recognized laboratory. The calibration records of the equipment are to be checked and preserved. The frequency of testing for 4 parameters PM10, PM2.5, SO₂, NO_x is fortnightly and other 8 parameters. Ozone O₃, Lead Pb, Carbon monoxide (CO), Ammonia (NH₃), Benzene (C₆H₆), Benzopyrene (BaP), Arsenic (As) and Nickel (Ni) to be tested once in six months at all 7 locations. The data is to be forwarded to EMG Dept. for onward transmission to Statutory Authorities as per the requirement.

Res. HOD (Chem)

- Meteorological parameters are to be maintained as per Annexure-V.

Res. HOD (EMG)

- Submission of record of details on calibration, CAAQMS availability and exception if any to Head (EMG), SRHQ and Head (EMG) CC on monthly basis as per Annexure-IV.

Res. HOD (EMG)

8.0 Review

This document shall be reviewed by HoP (Simhadri) on a two-yearly basis or as necessary.

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Annexure I - Checklist for monitoring of ambient air quality

S.N.	DESCRIPTION OF CHECKS	STATUS (YES/NO)	REMARKS
1.	Whether Station/project is having adequate numbers of AAQ stations as applicable to station/Project as per EC/CTO and applicable statutory norms/rules)		
2.	Whether all the stations are established in consultation with concerned authorities (APCB) as stipulated in EC/CTO/MoEF/CPCB direction.		
3.	Date of commissioning of equipment and system healthiness.		
4.	Whether data accessibility is made available to APCB/CPCB as required.		
5.	Whether availability of data is meeting as per requirement (>85%) and monitoring is being done in line with NAAQS standards 2009 and CPCB guidelines for AAQ monitoring.		
6.	Whether calibration is being done and records maintained as per document OGN/ENV/010.		
7.	Whether exceptions (Data availability/limit exceedance) are reported, and action taken as per document OGN /ENV/010		
8.	Whether O&M is being done by OEM/OES/third party or NTPC.		
9.	If not by NTPC, CAMC/AMC available.		
10.	Whether all analyzers are available and data available as per guidelines.		
11.	Whether data reporting is being done as per required formats in document ENV/010 and as required by APCB/CPCB/other regulators for station.		

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**ANNEXURE II - National Ambient Air Quality Standards, Central Pollution Control Board,
New Delhi, 18th November, 2009**

S.No	Pollutant	Time Weighted	Concentration in Ambient Air		
		Average	Industrial, Residential, Rural and other Area	Ecologically Sensitive area (notified By Central Government)	Method of Measurement
1	Sulphur	Annual*	50	20	-Improved west and Gaeke method
	Dioxide (SO ₂), µg/m ³	24 hours**	80	80	-Ultraviolet fluorescence
2	Nitrogen	Annual*	40	30	-Modified Jacob & Hochheiser (Na-Arsenite)
	Dioxide (NO ₂), µg/m ³	24 hours**	80	80	-Chemiluminescence
3	Particulate Matter(Size less than 10 µm) or PM ₁₀	Annual*	60	60	-Gravimetric
	µg/m ³	24 hours**	100	100	-TOEM -Beta Attenuation
4	Particulate Matter(Size less than 2.5 µm) or PM _{2.5}	Annual*	40	40	-Gravimetric
	µg/m ³	24 hours**	60	60	-TOEM -Beta Attenuation
5	Ozone(O ₃)	8 hours**	100	100	-UV photometric
	µg/m ³	1 hours**	180	180	- Chemiluminescence

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					-Chemical method
6	Lead (Pb) $\mu\text{g}/\text{m}^3$	Annual* 24 hours**	0.50 1.0	0.50 1.0	AAS/ICP method after sampling on EPM 2000 or equivalent filter paper -ED-XRF using Teflon filter
7	Carbon Monoxide (CO) mg/m^3	Annual* 24 hours**	02 04	02 04	-Non Dispersive Infra Red (NDIR) spectroscopy
8	Ammonia (NH ₃) $\mu\text{g}/\text{m}^3$	Annual* 24 hours**	100 400	100 400	- Chemiluminescence -Indophenol blue method
9	Benzene(C ₆ H ₆) $\mu\text{g}/\text{m}^3$	Annual*	05	05	-Gas chromatography based continuous analyser -Adsorption and Desorption followed by GC analysis
10	Benzo(α),Pyrene (BaP)- Particulate Phase only,	Annual*	01	01	-Solvent extraction followed by HPLC/GC analysis
11	Arsenic(As), $\mu\text{g}/\text{m}^3$	Annual*	06	06	-AAS/ICP method after sampling on EPM 2000
12	Nickel (Ni) ng/m^3	Annual*	20	20	-AAS/ICP method after sampling on EPM 2000

Annual arithmetic mean of minimum 104 measurements in a year at a particular site taken twice a week 24 hourly at uniform intervals.

** 24 hourly or 08 hourly or 01 hourly monitored values, as applicable, shall be complied with 98% of the time in a year.2% of the time; they may exceed the limits but not on two consecutive days of monitoring.

Note- - Whenever and wherever monitoring results on two consecutive days of monitoring exceed the limits specified above for the respective category, it shall be considered adequate reason to institute regular or continuous monitoring and further investigation.

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ANNEXURE III – CPCB Protocols For Data Transmission

CPCB IT Division

Protocol for data transmission from CAAQM stations existing as on date

1. Data Format

- Data file on real time basis having 15 minutes average values in a prescribed format attached at Annexure-I should be generated at the station for which Instrument Supplier is responsible.
- File should be updated in every 15 minutes
- Data intervals like 00:15,00:30,00:45,01:00 should be fixed at the station
- Station file name should be exactly as the name of the station to be displayed on the web portal i.e Sanathnagar, Nehrunagar. Here precaution is to be taken that no space between should be given or no special characters should be used.
- File should be recorded in a folder **C:\data\sanathnagardata.txt**
- File should allow data appending sequentially.
- File appending should continue subject to max 97 lines. First in First out mechanism shall be followed in keeping file size to 97 lines.
- Date of last file record appended in the file should be recorded and data afterwards be placed in the data file.
- Hence, in the specified folder C:\Data\ there will be a single file which will keep appending as per format attached
- Duplicate entry of any data should not be made in the file
- System should have capability to create previous record data file for which user will give the date. This is required to have lost data makeup in the final database, if any.

2. Data Mapping

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- Protocol for each parameter is fixed as below:
- 15 minutes average value will be provided by the operator of the CAAQMS
- Each SPB will have the parameter as mentioned in the table only. Not even a small gap or space is provided other than the mentioned table is acceptable.

3. Standard Parameter Naming Protocol and Conversion factors Table

Parameters Name	Parameter Abbreviations	Unit	Conversion factors at 25°C
Rack Temperature	Temp	°C	--
Carbon Monoxide	CO	mg/m ³	1 ppm=1.145 mg/m ³
Sulphur Dioxide	SO ₂	µg/m ³	1 ppb=2.62 µg/m ³
Nitric Oxide	NO	µg/m ³	1 ppb=1.23 µg/m ³
Nitrogen Dioxide	NO ₂	µg/m ³	1 ppb=1.88 µg/m ³
Oxides of Nitrogen	NO _x	ppb	--
Ozone	Ozone	µg/m ³	1 ppb=1.96 µg/ m ³
Particulate matter less than 10 micron size	PM ₁₀	µg/m ³	---
Wind Speed	WS	m/s	
Wind Direction	WD	deg	
Ambient Temperature	AT	°C	
Relative Humidity	RH	%	
Barometric Pressure	BP	mm/Hg	
Solar Radiation	SR	W/mt ²	
Rain Fall	RF	mm	
Vertical Wind Speed	VWS	degree	
Particulate matter less than 2.5 micron size	PM _{2.5}	µg/m ³	
Benzene	Benzene	µg/m ³	1 ppb=3.19 µg/ m ³
Toluene	Toluene	µg/m ³	1 ppb=3.77 µg/ m ³
Xylene	Xylene	µg/m ³	1 ppb=4.34 µg/ m ³
Ethyl Benzene	Eth-Benzene		1 ppb=4.34 µg/ m ³

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M+P_Xylene	MP-Xylene		1 ppb=4.34 $\mu\text{g}/\text{m}^3$
Methane		$\mu\text{g}/\text{m}^3$	1 ppb=0.65 $\mu\text{g}/\text{m}^3$
Ammonia		$\mu\text{g}/\text{m}^3$	1 ppb=0.70 $\mu\text{g}/\text{m}^3$
Formaldehyde		$\mu\text{g}/\text{m}^3$	1 ppb=1.23 $\mu\text{g}/\text{m}^3$
Mercury		$\mu\text{g}/\text{m}^3$	1 ppb=8.20 $\mu\text{g}/\text{m}^3$

Note : Any other parameter can be added with the prior approval of IT Division ONLY

4. Internet Connectivity

- Internet connectivity should be available on 24x7 basis for data transmission with an uptime of 99.9%. For this purpose every CAAQM should have two kinds of connection:
 - i) Leased Line Circuit of at least 01 Mbps capacity
 - ii) Broad Band connectivity through telephone line. Both facilities should be configured in ready to use condition. If possible auto failover should be created.

Note: Connectivity through Datacard is not acceptable except in any special circumstances, where both of these types of connectivities are not available. For such case CPCB IT Division shall be consulted before taking a final decision.

5. Other Information

1. Area Map showing station location
2. Latitude, Longitude and altitude of the station

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3. Photo of station along with nearby areas
4. One page write-up about the station activities in the vicinity of station including major pollution sources like nearby road, rail, restaurants, generator sets, etc.

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ANNEXURE-IV : Format for maintaining AAQ record offline monitoring

S.N o.	Date.	Location details of Ambient Air Quality Monitoring	Approx. Aerial Distance of CAAQMS from Stack (km)	Particulate Matter(Size less than 10 µm) or PM10 µg/m ³ on 24 hrly	Particulate Matter(Size less than 2.5 µm) or PM2.5 µg/m ³ on 24 hrly	SO ₂ (ug/ M ³) on 24 hrly basis	NO _x (ug/ M ³) on 24 hrly basis	CO (mg/ m ³)	Ozone(O ₃) µg/m ³ on 8 hrly basis	Remarks

Calibration/Validation Status of Continuous Ambient Air Quality Monitoring System (Monthly).

1. Date of Last Calibration of CAAQMS:
2. Deviation (%) recorded in last sampling for validation/verification:
3. Online percentage availability CAAQMS data in current Month and cumulative in current FY:
4. Reasons of less availability (<85 %) of AAQMS data.
5. Online data connectivity status to APCB/CPCB (Exceptions if any):
6. Name of the Person responsible for CAAQMS, Contact Number & E-mail ID:
7. Name of the person & his contact detail communicated to CPCB/APCB/CC:

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ANNEXURE-V : Format for maintaining Meteorological Parameters

MONITORING FREQUENCY : DAILY

S.No	Date	Time (Hrs.)	Temperature		RH(%) *	Wind Direction *	Wind Speed (m/s) *	Rainfall (24 hrs. basis)	Solar Flux (W/m 2)	Barometric pressure (mm Hg)	Remarks
			Max. °C	Min. °C				(mm)			